

**KALAMAZOO BATTLE CREEK, MI, USA**WMO: **726357**Lat: **42.235N**Lon: **85.552W**Elev: **868**StdP: **14.24**Time Zone: **-5.00 (NAE)**Period: **98-19**WBAN: **94815****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 1.5        | 7.0 | -5.6                          | 4.2 | 3.7  | -0.4 | 5.6 | 9.1  | 24.3                  | 26.6 | 21.9 | 24.9 | 6.5                   | 220  | 0.515 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.9                   | 90.1            | 73.1 | 87.6 | 71.8 | 84.4 | 70.2 | 76.1                | 85.4 | 74.5 | 83.3 | 72.9 | 81.3 | 10.4                 | 220  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 73.0                            | 126.3 | 81.6 | 71.8 | 121.4 | 80.2 | 70.1 | 114.5 | 78.2 | 40.0          | 85.5 | 38.5 | 83.3 | 37.0 | 81.2 | 82.2              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |    |    |
| (4) 21.7          | 19.0 | 17.2 | -5.2                       | 94.1 | 7.7                | 2.9 | -10.7                                              | 96.2 | -15.2      | 97.9 | -19.5      | 99.5 | -25.1      | 101.7 |    |    |
| (5)               |      |      | -5.7                       | 78.8 | 7.5                | 1.9 | -11.1                                              | 80.2 | -15.5      | 81.3 | -19.7      | 82.4 | -25.1      | 83.8  |    |    |

**Monthly Climatic Design Conditions**

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 50.0        | 24.8  | 27.1  | 36.6  | 49.1  | 60.2  | 69.1  | 72.9  | 71.0  | 64.3  | 52.1  | 40.8  | 30.3  |  |  |
|                                                                        | DBStd       | 19.09       | 10.86 | 10.46 | 11.27 | 9.22  | 8.61  | 6.79  | 5.67  | 5.16  | 7.64  | 8.64  | 9.00  | 9.91  |  |  |
|                                                                        | HDD50       | 2997        | 782   | 642   | 442   | 125   | 14    | 0     | 0     | 0     | 1     | 78    | 300   | 613   |  |  |
|                                                                        | HDD65       | 6237        | 1246  | 1060  | 882   | 482   | 201   | 35    | 6     | 10    | 104   | 410   | 727   | 1074  |  |  |
|                                                                        | CDD50       | 2993        | 1     | 2     | 26    | 98    | 329   | 574   | 709   | 652   | 431   | 143   | 24    | 4     |  |  |
|                                                                        | CDD65       | 758         | 0     | 0     | 2     | 5     | 51    | 159   | 250   | 197   | 84    | 10    | 0     | 0     |  |  |
|                                                                        | CDH74       | 6591        | 0     | 0     | 16    | 58    | 490   | 1387  | 2306  | 1542  | 712   | 80    | 0     | 0     |  |  |
|                                                                        | CDH80       | 2004        | 0     | 0     | 2     | 7     | 127   | 447   | 800   | 409   | 200   | 12    | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 7.7         | 9.3   | 9.0   | 8.9   | 9.0   | 7.5   | 6.7   | 6.2   | 5.7   | 6.0   | 7.4   | 8.5   | 8.9   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 36.4        | 2.0   | 1.6   | 2.4   | 3.4   | 3.3   | 3.7   | 3.8   | 3.5   | 3.9   | 2.9   | 2.9   | 2.7   |  |  |
|                                                                        | PrecMax     | 46.9        | 4.6   | 3.8   | 4.0   | 5.3   | 6.3   | 8.3   | 8.0   | 11.2  | 10.6  | 7.0   | 6.5   | 5.1   |  |  |
|                                                                        | PrecMin     | 26.9        | 0.6   | 0.0   | 0.2   | 0.8   | 0.6   | 0.7   | 0.7   | 0.6   | 0.0   | 0.9   | 0.4   | 0.6   |  |  |
|                                                                        | PrecStd     | 4.5         | 1.2   | 1.1   | 1.1   | 1.2   | 1.5   | 2.1   | 1.8   | 2.3   | 2.4   | 1.3   | 1.5   | 1.3   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 55.3  | 61.1  | 75.7  | 80.6  | 88.1  | 92.3  | 94.8  | 90.8  | 89.9  | 81.9  | 68.9  | 60.6  |  |  |
|                                                                        |             | MCWB        | 52.5  | 52.6  | 63.0  | 62.3  | 69.5  | 73.7  | 75.5  | 73.8  | 71.5  | 69.3  | 57.7  | 56.9  |  |  |
|                                                                        | 2%          | DB          | 49.3  | 52.5  | 67.2  | 74.5  | 83.5  | 88.5  | 90.4  | 87.4  | 85.7  | 75.4  | 63.1  | 54.4  |  |  |
|                                                                        |             | MCWB        | 45.6  | 47.0  | 57.5  | 59.6  | 67.1  | 71.4  | 74.2  | 73.1  | 71.0  | 63.3  | 54.9  | 51.2  |  |  |
|                                                                        | 5%          | DB          | 43.4  | 45.7  | 60.7  | 70.2  | 79.6  | 85.2  | 87.7  | 84.2  | 81.5  | 71.0  | 58.9  | 48.1  |  |  |
|                                                                        |             | MCWB        | 40.0  | 41.0  | 52.6  | 56.7  | 65.3  | 69.7  | 72.3  | 70.7  | 67.9  | 61.8  | 52.2  | 44.5  |  |  |
|                                                                        | 10%         | DB          | 39.0  | 41.0  | 54.4  | 65.3  | 75.3  | 81.8  | 84.2  | 81.8  | 78.1  | 66.3  | 54.7  | 43.3  |  |  |
|                                                                        |             | MCWB        | 36.4  | 37.3  | 47.4  | 53.8  | 63.5  | 68.6  | 70.7  | 69.6  | 66.0  | 59.1  | 48.9  | 40.0  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 53.1  | 55.0  | 64.1  | 66.2  | 72.7  | 76.4  | 78.8  | 77.4  | 75.2  | 71.0  | 60.1  | 57.4  |  |  |
|                                                                        |             | MCDB        | 55.1  | 58.3  | 75.6  | 75.7  | 83.3  | 87.5  | 90.3  | 86.6  | 85.5  | 80.0  | 65.1  | 60.6  |  |  |
|                                                                        | 2%          | WB          | 46.2  | 47.6  | 59.0  | 62.0  | 69.8  | 74.2  | 76.7  | 75.2  | 72.7  | 66.9  | 56.8  | 51.2  |  |  |
|                                                                        |             | MCDB        | 48.3  | 51.8  | 65.5  | 70.5  | 79.6  | 84.3  | 86.4  | 83.0  | 81.1  | 71.9  | 61.7  | 53.8  |  |  |
|                                                                        | 5%          | WB          | 40.4  | 41.8  | 54.0  | 58.8  | 67.7  | 72.4  | 75.0  | 73.4  | 70.5  | 63.2  | 53.2  | 45.1  |  |  |
|                                                                        |             | MCDB        | 43.2  | 45.2  | 59.6  | 67.4  | 76.2  | 81.8  | 83.9  | 81.1  | 77.8  | 68.5  | 58.1  | 47.3  |  |  |
|                                                                        | 10%         | WB          | 36.5  | 37.4  | 47.7  | 55.3  | 65.3  | 70.6  | 73.2  | 71.8  | 68.3  | 59.7  | 49.5  | 40.2  |  |  |
|                                                                        |             | MCDB        | 39.0  | 40.9  | 53.3  | 64.2  | 73.2  | 79.1  | 81.5  | 79.0  | 75.4  | 65.9  | 54.8  | 43.3  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 12.5  | 14.0  | 17.1  | 20.3  | 20.0  | 19.4  | 19.9  | 19.2  | 20.3  | 17.8  | 14.4  | 11.6  |  |  |
|                                                                        |             | MCDBR       | 15.9  | 19.3  | 24.7  | 25.9  | 24.1  | 22.3  | 22.1  | 21.5  | 23.2  | 23.4  | 20.3  | 16.9  |  |  |
|                                                                        | 5% WB       | MCWBR       | 13.6  | 15.1  | 16.2  | 14.8  | 12.3  | 9.7   | 9.4   | 9.5   | 10.7  | 13.5  | 14.6  | 14.5  |  |  |
|                                                                        |             | MCWBR       | 14.8  | 18.4  | 22.0  | 22.5  | 20.5  | 18.8  | 18.7  | 17.3  | 19.0  | 19.7  | 18.2  | 15.7  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.296 | 0.353 | 0.350 | 0.389 | 0.421 | 0.439 | 0.431 | 0.437 | 0.394 | 0.354 | 0.319 | 0.302 |  |  |
|                                                                        |             | taud        | 2.366 | 2.177 | 2.339 | 2.254 | 2.210 | 2.201 | 2.229 | 2.209 | 2.307 | 2.419 | 2.467 | 2.401 |  |  |
|                                                                        | Ebn at Noon |             | 268   | 264   | 283   | 279   | 271   | 266   | 267   | 261   | 265   | 264   | 259   | 255   |  |  |
|                                                                        |             | Edh at Noon | 26    | 37    | 36    | 42    | 45    | 45    | 43    | 43    | 36    | 29    | 23    | 23    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 432   | 697   | 1112  | 1472  | 1703  | 1928  | 1957  | 1686  | 1362  | 866   | 532   | 358   |  |  |
|                                                                        | RadStd      |             | 41    | 100   | 111   | 137   | 167   | 127   | 89    | 92    | 123   | 93    | 76    | 44    |  |  |

**Historical Trends**

|                              |       | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) Station Only            | DBAvg | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (47) Regional (65 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page